

Other popular IDE features are intended to assist developers in organising their workflow and problem-solving. IDEs parse code as it is written, identifying problems caused by a human mistake in real-time. Because a single graphical user interface represents utilities, developers can carry out activities without switching between apps. Most IDEs also include syntax highlighting, which employs visual clues to discern grammar in the text editor. Class and object browsers, and class hierarchy diagrams for specific languages are also included in several IDEs.

Today, most enterprise development teams choose a pre-configured IDE best suited to their specific use case. So the question is not whether to use an IDE, but which one to use.

The benefits of using an IDE

When writing a program, using an IDE will save you a lot of time and work. Among the benefits are:

- **Less time and effort:** An IDE's fundamental aim is to make development faster and easier. Its tools and features are designed to assist you in organising resources, avoiding mistakes, and providing shortcuts.
- **Enforce project or corporate standards:** A group of programmers will adhere to a traditional manner of doing things just by working in the same development environment. Bars can be enforced further if the IDE provides predefined templates, or code libraries are shared across multiple team members/teams working on the same project.
- **Project management:** This can be done in two ways. First, many IDEs provide documentation features that either automate developer comments or force developers to submit comments in various sections. Second, having a visual representation of resources should

make it much easier to understand how an application is laid out rather than traversing the file system for hidden files in it.

Features of IDEs

There are several technical and business applications for IDEs, which means numerous proprietary and open source IDE solutions are already on the market. The essential distinguishing features amongst IDEs are listed below.

The number of languages supported: Some IDEs are dedicated to a single language and hence better suited to a specific programming paradigm. IntelliJ, for example, is best known as a Java IDE. Other IDEs, such as the Eclipse IDE, offer a wide range of languages in one package, such as Java, XML, Python, and others.

Operating system(s) supported: A developer's operating system will limit which IDEs are viable (unless an IDE is cloud based). If the application created is intended for an end user with a particular operating system (such as Android or iOS), this may be an additional constraint.

Automation features: While most IDEs offer the three essential functions of a text editor, build automation, and debugger, many also enable refactoring, code search, and continuous integration and deployment (CI/CD) tools.

Impact on system performance: If a developer wants to run other memory-intensive apps concurrently, the memory footprint may be vital to consider.

Plugins and extensions: Some IDEs feature the ability to adapt workflows to meet needs and preferences.

Popular open source IDEs for Windows

NetBeans

This is a free and open source IDE that is great for tweaking existing projects

or creating new ones. NetBeans has a simple drag-and-drop interface and a plethora of helpful project templates. It is typically used to construct Java applications, although it can get packages or bundles that support other languages. C, FORTRAN, C++, C++11, HTML 5, Java, PHP, and other languages are supported.

Features

- NetBeans has an easy-to-use, drag-and-drop interface that simplifies the software development process.
- It includes both static and dynamic libraries for use in software development.
- GNU debuggers with many sessions can be readily integrated to provide code assistance.
- NetBeans allows its users to develop software from a remote location.
- It works with Windows, Linux, OS X, and Solaris.
- It can read and write Fortran and Assembler files.
- Qt Toolkit is supported.
- It works with various compilers, including Cygwin, Clang/LLVM, MinGW, GNU, and Oracle Solaris Studio.

Limitations

Some NetBeans users have noted that it takes a lot of RAM. Therefore it may function sluggishly on machines with insufficient memory.

Code::Blocks

Code::Blocks is a well-known open source IDE for developers working on Windows, Linux, and Mac. It is a highly customisable IDE that runs consistently across all platforms, ideal for developers who regularly swap workspaces across systems. The plugin framework supported by Code::Blocks allows users to customise it to their specific needs and requirements.